ADDRESSING MODES

Instruction Format

"The assembler processes an instruction, and it converts the instruction from its mnemonic form to a standard machine-language (binary) format called an "instruction format"

Components of Instruction format

- Operation code
  - Specifies operations to be performed.
- Address field
  - Designates a memory address or a processor register.
- Mode
  - The way the operand or the effective address is determined.

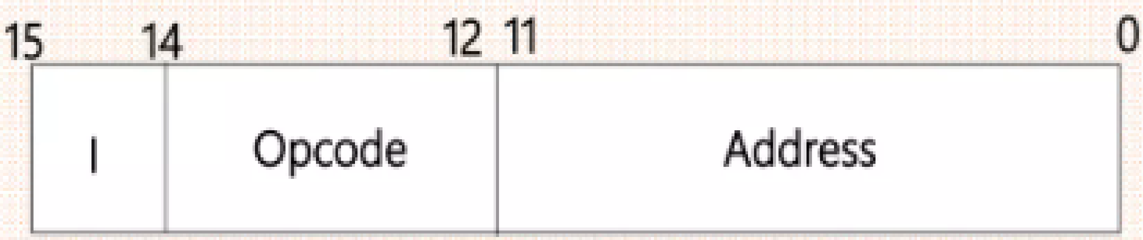
Instruction format

In an instruction format:

- First 12 bits(0-11) specify an address
- Next 3 bits specify operation code(opcode)
- Left most bit specify the addressing mode I

I=0 for direct address

I=1 for indirect address



Types of Instruction Format

- Three address instruction
  - ADD R1, A, B //R1<-M[A]+M[B]
- Two address instruction
  - MOV R1, A //R1<-M[A]
- One address instruction
  - LOAD A //AC<-M[A]
- Zero address instruction
  - PUSH A //TOS<-A

Addressing Modes

- The way the operands are chosen during program execution is dependent on the addressing mode of the instruction.
- The addressing mode specify a rule for interpreting or the modifying the address field of the instruction before the operand is actually referenced.

Opcode	Mode	Address

Figure Instruction format with mode field

Types of Addressing modes

- Implied Mode
- Immediate Mode
- Register Mode
- Register Indirect Mode
- Autoincrement or Autodecrement Mode
- Direct Address Mode
- Indirect Address Mode
- Relative Address Mode
- Indexed Addressing Mode
- Base Register Addressing Mode

Addressing Modes	Applications
Immediate Addressing Mode	Initialize register to a constant value.
Direct Addressing Mode and Register Direct Addressing Mode	Helps access static data and implement variables.
Indirect Addressing Mode and Register Indirect Addressing Mode	Helps implement pointers and pass arrays as parameters.
Relative Addressing Mode	Helps in program relocation at runtime. And in changing the sequence of instruction during execution.
Index Addressing Mode	Helps in the array and record implementation.
Base Register Addressing Mode	Helps in writing relocatable code and handling recursive procedures.
Auto- Increment Addressing Mode and Auto- Decrement Addressing Mode	Helps implement loops and stacks.

Implied Mode

- The operand is hidden/fixed inside the instruction.
- Zero Address instructions in a stack organized computer are implied mode instructions.

Complement Accumulator CMA

(Here accumulator A is implied by the instruction)

Complement Carry Flag CMC

(Here Flags register is implied by the instruction)

Set Carry Flag STC

(Here Flags register is implied by the instruction)

Immediate Mode

- In this mode the operand is specified in instruction itself.
- Operand itself is provided in the instruction rather than its address.

Move Immediate

MVI A , 15h A ← 15h Here 15h is the immediate operand

Add Immediate

ADI 3Eh A ← A + 3Eh Here 3Eh is the immediate operand

Register Mode

- The operand is specified with in one of the processor register.
- Instruction specifies the register in which the operand is stored.

Move

MOV C , A C ← A Here A is the operand specified in register

Add

ADD B A ← A + B Here B is the operand specified in register

Register Indirect Mode

- The instruction specifies the register in which the memory address of operand is placed.
- It do not specify the operand itself but its location with in the memory where operand is placed.

Move

MOV A , M A ← [[H][L]]

It moves the data from memory location specified by HL register pair to A.

Direct Addressing Mode

- The instruction specifies the direct address of the operand.
- The memory address is specified where the actual operand is.

Load Accumulator

LDA 2805h A ← [2805]

It loads the data from memory location 2805 to A.

Store Accumulator

STA2803h [2803] ← A

It stores the data from A to memory location 2803.

Thanks A Lot....!!